



Predicting the spread of COVID-19 with a machine learning technique and multiplicative calculus

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Abstract

This paper aims to generate a universal well-fitted mathematical model to aid global representation of the spread of the coronavirus (COVID-19) disease. The model aims to identify the importance of the measures to be taken in order to stop the spread of the virus. It describes the diffusion of the virus in normal life with and without precaution. It is a data-driven parametric dependent function, for which the parameters are extracted from the data and the exponential function derived using multiplicative calculus. The results of the proposed model are compared to real recorded data from different countries and the performance of this model is investigated using error analysis theory. All statistics, collected data, etc., included in this study were extracted from official website of the World Health Organization (WHO). Therefore, the obtained results demonstrate its applicability and efficiency.

Keywords Multiplicative least square method · COVID-19 model · Multiplicative data fitting · Simulation

1 Introduction

Coronaviruses (CoV) belong to the family of zoonoses virus. These are transmitted by animals to humans and cause symptoms such as common cold or more serious illnesses such as Eastern respiratory syndrome (MERS). The latter is transmitted from dromedaries to humans, or severe acute respiratory syndrome (SARS), which is transmitted by civets to humans. Several types of known coronaviruses that have not yet infected humans circulate among some animals. Disease due to the coronavirus appeared in December 2019, commonly known as “COVID-19,” is a very contagious disease and caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The SARSCoV-2 is a new strain of coronavirus which had not yet been identified in humans, where respiratory symptoms, fever and cough are common signs of infection of the new coronavirus pandemic. In some cases, this disease can cause pneumonia, severe acute breathing problems, kidney failure and even death. Researchers from several fields have been working to prevent or minimize the rate of some infections in. In particular, mathematicians have investigated the evaluation of relevant nonlinear dynamics of problems related to infections, such as epidemics and pandemics. Starting from 1990, mathematicians, biologists and researchers working in medical science have been working hard to get more information about how epidemics spread and

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how to prevent them from growing in the community. In the area of preventing diseases, mathematical modeling plays a considerable role and great contribution using mathematical approaches as well as simulation. Such approaches may then be used to come up with some strategies and/or techniques to forecast or manage infectious diseases spreading.

On the other hand, multiplicative calculus presents a wide range of applications in science and engineering. After introductory paper Bashirov et al. (2008), many research papers related to multiplicative calculus have been followed to improve problem solutions in different applications. The papers Bilgehan (2015), Bashirov et al. (2011), Filip and Piatecki (2014) and Özyapıcı et al. (2017) are suggested for basic applications of the multiplicative calculus. Another important mathematical area of multiplicative calculus is numerical analysis. Many numerical methods in numerical analysis have been introduced in recent years. The paper Özyapıcı et al. (2014) has been compared with similar minimization methods and has proven its superiority. The papers Özyapıcı et al. (2016), Özyapıcı (2020), via numerical analysis, also showed that multiplicative numerical methods can be applied successfully while classical methods fail. In the paper Bilgehan (2015), the optimal representation of linear, nonlinear type signals has been introduced based on the multiplicative least square method.

In the paper Özyapıcı and Mısırlı (2009), multiplicative interpolation methods were introduced and later these methods were effectively used to determine the learning curve in the paper Özyapıcı et al. (2017). The multiplicative differential equations are also effectively used in many applications, some important numerical methods for multiplicative differential equations were introduced in the papers Mısırlı and Gurefe (2011) and Rıza et al. (2009), respectively.

The methods and applications in all these papers have been constructed using some of the basic properties of multiplicative calculus, especially based on the definition of multiplicative derivative. In addition, in this study, it is thought that the spread of the coronavirus is exponential and the multiplicative-based representation of the spread of the coronavirus can represent an excellent approach to real-life examples. Therefore, the multiplicative least squares method which is the basis for the mathematical representation of the spread of COVID-19 is used in this paper. As in all other methods in the literature, the multiplicative least squares method has been revealed by using some features of the multiplicative calculus based on ordinary and partial multiplicative derivative. Therefore, since we will use the multiplicative derivative when introducing the multiplicative least-square method, we also need to give the definition of multiplicative derivative as follows:

Definition 1 (Bashirov et al. 2008) Let f be a positive function with respect to x where $x \in \text{dom}(f)$. If its classical

derivative at x exists, then multiplicative derivative of f can be easily written as

$$f^*(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h)}{f(x)} \right)^{\frac{1}{h}} = \exp \{ f'(x)/f(x) \} \\ = \exp \{ (\ln \circ f)'(x) \}, \quad (1)$$

where $(\ln \circ f)(x) = \ln f(x)$. Applying this derivative n times, we determine that if f is a positive function and its n th classical derivative at x exists, then $f^{*(n)}(x)$ exists as follows:

$$f^{*(n)}(x) = \exp \{ (\ln \circ f)^{(n)}(x) \} \quad (2)$$

Also, some important properties of multiplicative derivative are given in the following theorem:

Theorem 1 (Bashirov et al. 2008) If f and g be differentiable in the multiplicative sense, then cf , $f \cdot g$, $f + g$, f/g and f^h are differentiable in the multiplicative sense, and

$$\begin{aligned} (cf)^*(x) &= f^*(x), \\ (f \cdot g)^*(x) &= f^*(x)g^*(x), \\ (f + g)^*(x) &= f^*(x)^{\frac{f(x)}{f(x)+g(x)}} \cdot g^*(x)^{\frac{g(x)}{f(x)+g(x)}}, \\ (f/g)^*(x) &= f^*(x)/g^*(x), \\ (f^h)^*(x) &= f^*(x)^{h(x)} \cdot f(x)^{h'(x)}, \end{aligned} \quad (3)$$

where c is a positive constant, f and g are multiplicative differentiable and h is classical differentiable.

Theorem 2 (Multiplicative Chain Rule Bashirov et al. (2008)) If f is a function of two variables y and z with continuous partial *derivatives, and also y and z are differentiable functions on (a, b) such that $f(y(x), z(x))$ is defined for every $x \in (a, b)$, then

$$\frac{d^* f(y(x), z(x))}{dx} = f_y^*(y(x), z(x))y'(x) f_z^*(y(x), z(x))z'(x) \quad (4)$$

here partial *derivative of f in x is denoted by f_x^* or $\frac{\partial^* f}{\partial x}$ and similarly partial *derivative of f in y is denoted by f_y^* or $\frac{\partial^* f}{\partial y}$.

In addition, many works have been published in the literature regarding mathematical models describing the spread of COVID-19 and the solutions of the problems described by these models. Some of them are given in References (Danane et al. 2021, 2019; Habenom et al. 2011; Özyapıcı et al. 2020; Babaei et al. 2021; GAO 2020; Gao et al. 2020; Veerasha et al. 2020; Özköse and Yavuz 2021; Ikram et al. 2022; Naik et al. 2020; Statistics of World Health Organization XXX) and the references therein.

1.1 Multiplicative least square method (MLSM)

In the literature, it was seen that the MLSM was, respectively, applied to the exponential signal processing Ozyapici and Bilgehan (2016), the low noise measurement in electrical circuits Bilgehan et al. (2018) and the experimental data in electrical circuits and systems Ozyapici and Bilgehan (2016). The general formula of MLSM can be introduced as follows:

$$S = \prod_{i=1}^n \left[\left(\frac{y_i}{f(\mathbf{b}, x_i)} \right)^{\ln \left(\frac{y_i}{f(\mathbf{b}, x_i)} \right)} \right] \quad (5)$$

In many applications as well as the spread of coronavirus, the set of real data $y_i, \forall i$ depends on observation. The previous papers states that the data $y_i, \forall i$ act exponentially, and the MLSM method can be applied effectively. Hence, the multiplicative least square method applied to determine the accurate parameters of $\mathbf{b} = \{b_1, b_2, \dots, b_n\}$.

The evaluation of the exponential functions in multiplicative least square method handled as linear functions that requires less computational time and produce more accurate parameter values. The accuracy of the parameter values result in better representation of data fitting applications. Hence, parameters of $\mathbf{b}$ can be determined by using multiplicative least-square Eq. (5). Therefore, the derivative of S with respect to each parameter $\mathbf{b}_i$ in vector $\mathbf{b}$ should be expressed. The equation in the multiplicative calculus should be set to one.

The expression follows as:

$$\begin{aligned} \left(\frac{\partial S}{\partial b_i} \right)^* &= \exp \left\{ \frac{\partial}{\partial b_i} \sum \ln(S_i) \right\} \\ &= \exp \left\{ - \sum_{i=1}^n \ln \left(\frac{y_i}{f(\mathbf{b}, x_i)} \right) \frac{1}{f(\mathbf{b}, x_i)} \frac{\partial f(\mathbf{b}, x_i)}{\partial b_i} \right\} \quad (6) \\ &= 1. \end{aligned}$$

2 Exponential data fitting process of spread of the coronavirus using MLSM

The main purpose is to introduce the best mathematical model to fit data related to the spread of coronavirus obtained in different countries such as China, Iran and Italy. The introduced mathematical model is a parametric exponential-based function. The parameters are evaluated using the MLSM.

The exponential functions

$$E(x) = e^{-(ax+b)}, \quad a > 0, \quad (7)$$

$$N(x) = e^{-(ax+b)^2}, \quad a, b > 0, \quad (8)$$

$$R(x) = xe^{(-ax^2)}, \quad a > 0. \quad (9)$$

are close to well-known exponential distribution functions. The parameters a and b in Eqs. (7–9) can be derived by using MLSM without a need to normalize the observed data.

Since the behavior of the spread of coronavirus is very important to take some necessary precautions in time, its accurate mathematical representation becomes very important in analysis and arrange a reaction. The starting point of deriving a best fit function can be one of the exponential functions (7–9). The process may possess difficulty when the behavior of the data is not obvious, because the behavior of the virus can be a little different from a country to another. But we all agree that the spread of the coronavirus is exponential. Therefore, we propose to use a general model with the exponential functions (7–9)

$$f(\mathbf{b}, x) = x^{b_1} e^{b_2 x^2 + b_3 x + b_4}. \quad (10)$$

Equation (10) is introduced in the paper Bilgehan et al. (2018). As $b_1 \rightarrow 0$, the function (10) approaches to Gaussian function. Consequently, when $b_1, b_2 \rightarrow 0$, the function (10) reduces to the exponential function. Finally, when $b_1 \rightarrow 1, b_2 \rightarrow 0$, the function (10) reduces to Rayleigh function. Additionally, the MBM in (10) used effectively in the paper Bilgehan et al. (2018) to introduce accurate solution to modern electrical circuits and systems.

The function (10) is linear in the multiplicative calculus. Therefore, the method MLSM directly applies to the function (10) with the multiplicative derivatives 6. Hence, substituting (10) into (5) gives:

$$\begin{aligned} S &= \prod_{i=1}^n \left[\left(\frac{y_i}{x^{b_1} e^{b_2 x^2 + b_3 x + b_4}} \right)^{\ln \left(\frac{y_i}{x^{b_1} e^{b_2 x^2 + b_3 x + b_4}} \right)} \right] \\ &= \prod_{i=1}^n \exp \left\{ \ln \left(\frac{y_i}{x^{b_1} e^{b_2 x^2 + b_3 x + b_4}} \right)^2 \right\} \quad (11) \end{aligned}$$

Lets now use the definition of the partial multiplicative derivative as in (11) with respect to b_1, b_2, b_3 and b_4 . The next step is to set up four differential equations to seek solutions for the four unknown parameters.

$$\left(\frac{\partial S}{\partial b_1} \right)^* = \sum_{i=1}^n (\ln x_i) (\ln y_i - b_1 \ln x_i - b_2 x_i^2 - b_3 x_i - b_4) = 0, \quad (12)$$

$$\left(\frac{\partial S}{\partial b_2} \right)^* = \sum_{i=1}^n (x_i^2) (\ln y_i - b_1 \ln x_i - b_2 x_i^2 - b_3 x_i - b_4) = 0, \quad (13)$$

$$\left(\frac{\partial S}{\partial b_3} \right)^* = \sum_{i=1}^n (x_i) (\ln y_i - b_1 \ln x_i - b_2 x_i^2 - b_3 x_i - b_4) = 0, \quad (14)$$

$$\left(\frac{\partial S}{\partial b_4} \right)^* = \sum_{i=1}^n (1) (\ln y_i - b_1 \ln x_i - b_2 x_i^2 - b_3 x_i - b_4) = 0. \quad (15)$$

To find the unknown parameters $\mathbf{b} = [b_1, b_2, b_3, b_4]$, the system of linear Eqs. (12–15) should be solved. In the next section, the numerical results show that the generalized function (10) can accurately represent the spread of the coronavirus in the different countries. In the further application, the function (10) can also be used for the representation of the spread of the other viruses which behaves like coronavirus. The scientific research in this article indicates the applicability of the function (10) using the MLSM. All the mathematical representations based on countries China, Iran and Italy in the next section show that the function (10) is a global representation of the spread of coronavirus and similar micro-organisms.

3 Coronavirus spread in China

The virus has been discovered on someone at the Huanan seafood market in Wuhan. The virus is assumed to be transmitted from animals. The coronavirus disease has a high spread rate from person to person. Such a characteristic is a drastic situation for all countries around the world and requires investigation to discover the unknowns for such disease. The disease spread via droplets of body fluids. The droplets can be absorbed directly from infected people or by touching infected surfaces. Transmission is a great concern to control the spread rate of the disease. China took some important actions to reduce the spread of the virus. China has ordered Wuhan and twelve other cities under lockdown with almost sixty million people. The country took further actions as the number of confirmed cases increased. The aim of the restrictions on movements, closing borders and stopping flights intended to reduce the spread rate. This action believed to gain time for the recovery period. Another advantage of such a limitation believed to minimize the strain on health services by avoiding an excess number of infected people at any one time. This analysis produces a universal mathematical model for different countries and indicates the infection rate concerning various actions taken by different countries. China is an example to introduce a complete lockdown as a solution to the fast-spreading coronavirus. The number of infected people under complete lockdown recorded by the authorities is tabulated in Table 1.

Table 1 indicates the monthly infections in China. This article uses the immediately available data at a time of occurrence. The original data and the multiplicative coronavirus model are plotted in Fig. 1. The fitting model uses MATLAB software. The fitness of the multiplicative coronavirus model is confirmed using the minimum relative error analysis.

The introduced model in (10) is applied to estimate the parametric values of the recorded data in Table 1. The solution of the parameters $\mathbf{b} = [b_1, b_2, b_3, b_4]$ follows the

Table 1 Number of infected people in a month

Day	Infected	Day	Infected	Day	Infected
1	48315	11	75891	21	80754
2	55220	12	76288	22	80778
3	58761	13	76936	23	80793
4	63851	14	77150	24	80813
5	66492	15	77658	25	80821
6	68500	16	78064	26	80844
7	70548	17	78497	27	80860
8	72436	18	78824	28	80881
9	74185	19	79211	29	80894
10	75002	20	79824	30	

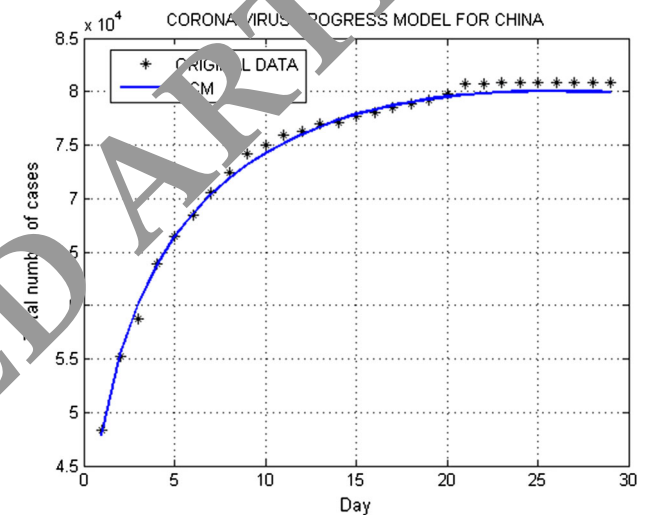


Fig. 1 Total number of infections per day

process described in Eqs. (12)–(15). The equations are formulated as:

$$10^3 * \begin{bmatrix} 0 & -26 & -1 & 0 \\ -26 & -4464 & -189 & -9 \\ -1 & -189 & -9 & 0 \\ 0 & -8 & -0 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} -802 \\ -96545 \\ -4901 \\ -324 \end{bmatrix}. \quad (16)$$

According to the system of linear Eq. (16) the parameters $\mathbf{b} = [b_1, b_2, b_3, b_4]$ are

$$b_1 = 0.2251, b_2 = 0.000014418, b_3 = -0.0088, b_4 = 10.7844.$$

Therefore, a new representation of spread of the coronavirus in China can be derived as

$$f(\mathbf{b}, x) = x^{0.2251} e^{-0.0088x + 10.7844}, \quad (17)$$

where the parameter b_2 is assumed to be zero.

Table 2 Relative error variations for spread of Covid-19 in China

Day	Error(Rel.e)	Day	Error (Rel.e)	Day	Error (Rel.e)
1	0.0098	11	0.0090	21	0.0130
2	0.0039	12	0.0034	22	0.0116
3	0.0245	13	0.0026	23	0.0105
4	0.0028	14	0.0025	24	0.0100
5	0.0019	15	0.0028	25	0.0097
6	0.0007	16	0.0033	26	0.0098
7	0.0028	17	0.0027	27	0.0103
8	0.0079	18	0.0026	28	0.0111
9	0.0139	19	0.0007	29	0.0121
10	0.0099	20	0.0037	30	

To show the superiority of the mathematical representation (17) of the spread of corona virus, the verification of the relative error analysis stated in Table 2 as

Where $\text{Rel.e} = \left| 1 - \frac{\text{the function (17)}}{\text{Original data}} \right|$. The error level of the introduced method is very low even with limited data points. Table 2 proves the new mathematical representation (17) to be suitable for the spread of coronavirus and similar other diseases following a similar spread rate. The success of the multiplicative representation is due to accurate parameter values.

China reacted very quickly, implementing some measures that would later set an example to the world. Table 1 shows the number of infected people when the outbreak occurred. At the end of the month, the rate of spread in China had almost come to a standstill. This can also be observed from the mathematical expression in Eq. (17).

4 Further MCM fitting

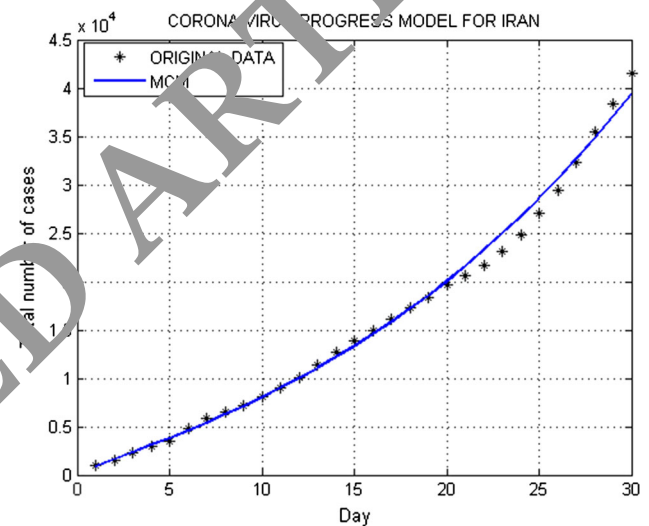
The purpose of this section is to prove the validity of the model in line with the measures taken in different countries. The two countries that implemented different measures than China are Iran and Italy. Hence, these countries are selected for the test of representative function (10).

4.1 Representation of the Coronavirus in Iran

Iran did not predict early a big domestic coronavirus outbreak and has also been criticized for not being transparent in updating the local situation to the rest of the world. Critics say the Iranian government should have taken tougher measures earlier on to tackle the spread of the coronavirus. It is not yet clear the date of an outbreak in Iran. For this reason, two major problems are said to have arisen. The first problem is that late precautions have been taken and the second problem is that international travels have continued. Due to

Table 3 Number of infected people in a month

Day	Infected	Day	Infected	Day	Infected
1	978	11	9000	21	20610
2	1501	12	10075	22	21638
3	2336	13	11364	23	23049
4	2922	14	12729	24	24811
5	3513	15	13938	25	27011
6	4747	16	14991	26	29406
7	5823	17	16169	27	32332
8	6566	18	17371	28	35408
9	7161	19	18477	29	38309
10	8042	20	19641	30	41495

**Fig. 2** Total number of infections per day

these two problems, the spread of the coronavirus has a very different course compared to China.

Table 3 indicates the number of infected people in the first month of the country Iran. This article uses the immediately available data at a time of occurrence. The original data and the multiplicative coronavirus model are plotted in Fig. 2. The fitting model uses MATLAB software. The fitness of the multiplicative coronavirus model confirmed using the minimum relative error analysis.

The parameters $\mathbf{b} = [b_1, b_2, b_3, b_4]$ can be evaluated by solving the following system of equations

$$10^3 * \begin{bmatrix} -0.2 & -29 & -1 & -0.1 \\ -29 & -5274 & -216 & -10 \\ -1 & -216 & -10 & -0.5 \\ -0.1 & -10 & -0.5 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} -721 \\ -95194 \\ -4579 \\ -280 \end{bmatrix} \quad (18)$$

should be solved. Consequently, the parameters based on (18) can be found as

$$b_1 = 0.8182, b_2 = 0, b_3 = 0.0344, b_4 = 6.7673.$$

Table 4 Relative error variations for the function (19) in Iran

Day	Error (Rel.e)	Day	Error (Rel.e)	Day	Error (Rel.e)
1	0.0804	11	0.0024	21	0.0478
2	0.0934	12	0.0048	22	0.0730
3	0.0132	13	0.0250	23	0.0812
4	0.0608	14	0.0428	24	0.0764
5	0.0962	15	0.0427	25	0.0578
6	0.0253	16	0.0289	26	0.0387
7	0.0671	17	0.0208	27	0.0084
8	0.0449	18	0.0109	28	0.0183
9	0.0019	19	0.0092	29	0.0335
10	0.0026	20	0.0206	30	0.0505

Therefore, a new representation of spread of corona virus in Iran can be derived as

$$f(\mathbf{b}, x) = x^{0.8182} e^{0.0344x+6.7673}. \quad (19)$$

To show the superiority of the mathematical representation (19) of the spread of corona virus, the verification of the relative error analysis will be given in Table 4 as

where $\text{Rel.e} = \left| 1 - \frac{\text{the function (19)}}{\text{Original data}} \right|$. The model function (10) with MLMS works well for the representation of the spread of Covid-19 in Iran. Consequently, the model function (19) can nearly reflect the real life situation of the spread of the Covid-19 which is justified by Table 4.

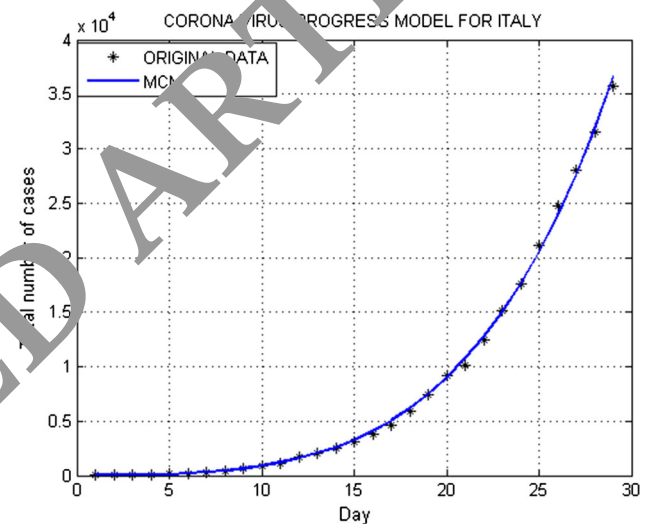
4.2 Representation of the Coronavirus in Italy

Italy is a popular tourist destination in the European Union. It is reported of two coronavirus (COVID-19) positive cases in Chinese tourists. Consequently, Italy became the country which has the highest cases in Europe. The criticism states the high number of coronavirus spread as the expanding air travel with China. This is following from the memorandum of understanding (signed in January 2020) between Italy and China. Although measures were taken in the north of the country at the beginning, no measures were taken in the rest of the country. The recorded infection cases proved wrong to underestimate the spread of coronavirus without taking serious measures. During the early times, the government took weak measures such as screening and suspending major community events. The measures are increased by closing the educational institutes. The Italian government lately took strong action by introducing a complete lockdown. The slowly taken measures are reflected in the recorded cases in Table 5.

Table 5 indicates the number of infected people in the first month of the Italy. This article use the immediate available data at a time of occurrence. The original data and the multi-

Table 5 Number of infected people in a month

Day	Infected	Day	Infected	Day	Infected
1	2	11	1128	21	10149
2	3	12	1694	22	12463
3	20	13	2036	23	15113
4	79	14	2502	24	17660
5	150	15	3089	25	21157
6	227	16	3858	26	27747
7	320	17	4636	27	27980
8	445	18	5887	28	31506
9	650	19	7377	29	35713
10	888	20	9172	30	

**Fig. 3** Total number of infections per day

plicative coronavirus model are plotted in Fig. 3. The fitting model use MATLAB software as well. The fitness of the multiplicative coronavirus model confirmed using the minimum relative error analysis as similar as the previous countries China and Iran.

The parameters $\mathbf{b} = [b_1, b_2, b_3, b_4]$ of the function (10) based on MLMS for the spread of Covid-19 of Italy can be evaluated by the following system of the equations

$$10^3 * \begin{bmatrix} -0.2 & -26 & -1.3 & 0 \\ -26 & -4464 & -189 & -14 \\ -1.3 & -189 & -9 & -0 \\ -0.1 & -9 & -0.4 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} -591 \\ -79901 \\ -3821 \\ -215 \end{bmatrix}. \quad (20)$$

The calculated parameters are:

$$b_1 = 2.6939, b_2 = 0, b_3 = 0.0488, b_4 = 0.1354$$

Substituting the parameters:

$$f(\mathbf{b}, x) = x^{2.6939} e^{0.0488x+0.1354}. \quad (21)$$

Table 6 Relative error variations for the function (21) in Italy

Day	Error(Rel.e)	Day	Error (Rel.e)	Day	Error (Rel.e)
1	0.4012	11	0.0620	21	0.0550
2	0.7014	12	0.0651	22	0.0185
3	0.2633	13	0.0094	23	0.0098
4	0.2740	14	0.0488	24	0.0061
5	0.2705	15	0.0700	25	0.0314
6	0.1761	16	0.0661	26	0.0375
7	0.0741	17	0.0925	27	0.0143
8	0.0021	18	0.0503	28	0.0097
9	0.0187	19	0.0137	29	0.0240
10	0.0022	20	0.0212	30	

The set of relative errors given in Table 6 shows the spread of the coronavirus in Italy. The error level is very low that shows a perfect representation by the model function (21) via the process of MLSM.

Where $\text{Rel.e} = \left| 1 - \frac{\text{the function (21)}}{\text{Original data}} \right|$. After the mathematical representations of the spread of coronavirus, we can easily understand the importance of the precautions considered by almost every country around the world. We have seen that immediate actions are very important and might reduce the number of deaths. The high rate of the spread of coronavirus in Italy mainly depends on the late taken actions. According to the mathematical representation (21), we can easily understand the behavior of the spread of the Covid-19 virus. For example, the nonlinear equation

$$f(\mathbf{b}, x) = x^{2.6939} e^{0.0488x+0.1354} = 10,000,000 \quad (22)$$

shows the number of days taken to spread the infection to 10 million people. The paper [?] suggests an alternative method as a solution to Eq. (22). The paper [?] uses multiplicative Halley method to solve Eq. (22). Consequently, an infected number of people can exceed ten million people in only 4 days from the beginning of the spread of the virus. The number of infections is well above the maximum limits than any country can handle. The mathematical representation of MCM yields accurate values scientifically rather than rough estimation. Graphical representation of the mathematical models based on data of newly pandemic COVID 19 are given in Figs. 1, 2 and 3. These representations are based on the data of Tables 1, 3 and 5 for China, Iran and Italy, respectively. The data of the tables are given from official website of the World Health Organization (WHO) Babaei et al. (2021). As can be seen from these results, considering the new pandemic COVID19 data, it will be possible to make predictions about the spread of the disease with minimum error.

5 Conclusion

In this paper, we have presented a new method (MCM) for the coronavirus spread. The success of the introduced method is due to the theory of multiplicative calculus. Through the real universal examples, the performance of MCM examined. MCM validity has been proved by using the error analysis method.

The MCM has introduced a new approach to represent the spread of the coronavirus. The MCM improved through the error analysis method to provide an accurate representation of the coronavirus spread. The MCM is universal and it represents accurately the global spread of the coronavirus disease. The selection of various applications proves that the model is not affected by the measures taken by different countries. The MCM can be used for different diseases showing similar symptoms. Close application of MCM will be advantageous in determining more accurate results in the very early stage of the disease. The MCM application to the Chinese example represents the infection spread rate with controllable factors such as a lockdown. The MCM also represents the infection rate without any precautions. The introduced model accurately represents the universal infection rate of the coronavirus.

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Data availability statement Enquiries about data availability should be directed to the authors.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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